

Title

Predictive Analytics for Brasileirão: A Data Driven Business Strategy for Sports Intelligence
Final Data Science Assignment
Sports intelligence prototype for Brasileirão

Business Problem

Clubs, media, and fantasy products need match-level insights.
Historical data can support probability-based previews.
The goal is to predict home win, away win, or draw.
The model is a decision-support signal.

Market Opportunity

Sports data products turn raw results into decisions.
Clubs can use predictions for opponent previews.
Media platforms can enrich match storytelling.
Fantasy products can improve engagement.

Dataset Overview

Local source: IA_FUTEBOL_DADOS.zip.

Standings data covers 2003-2022.

Match data contains fixtures, scores, teams, venues, and winners.

Goal data is available for deeper future work.

Exploratory Data Analysis

Champion and runner-up counts show historical dominance.

Average points summarize long-term strength.

EDA supports both modeling and business storytelling.

Charts help explain why some clubs are favored.

Machine Learning Approach

Built a standings-position baseline.

Built a match-level classifier for final use.

Features: home team, away team, and title counts.

Target classes: draw, home win, away win.

Model Evaluation

Compared SVC, kNN, Decision Tree, and XGBoost.

SVC remains the final model from the original workflow.

Confusion matrices show where predictions succeed or fail.

Draws are the hardest class to separate.

Key Insights

Historical strength is a useful but limited signal.

Match probabilities are easier to productize than raw labels.

The model can support previews and dashboards.

Predictions should be presented with uncertainty.

Business Strategy

Launch a Brasileirão Sports Intelligence Dashboard.

Club module: opponent previews.

Media module: probability-based match cards.

Fantasy module: fixture difficulty signals.

Start simple before adding richer data.

Limitations

Small feature set.

No player, injury, transfer, or tactical data.

Simple team encoding limits model depth.

Historical success may not reflect current form.

Not a standalone decision system.

Next Steps

Add recent form and rolling performance features.

Include player availability and transfer data.

Test stronger encodings for teams.

Track model performance season by season.

Package predictions in a simple dashboard.

References

Kaggle standings and match datasets.
CBF historical competition information.
pandas, NumPy, scikit-learn.
seaborn, Matplotlib, XGBoost.